Titanic Dataset Analysis Report

1. Title

Titanic Dataset Cleaning and Exploratory Data Analysis (EDA)

2. Introduction

This project was completed using the Titanic dataset to perform data cleaning and exploratory data analysis. The goal of the analysis was to understand the structure of the dataset, prepare the data for analysis by handling missing values and other issues, and then use visualizations to identify important patterns related to passenger survival.

The Titanic dataset is one of the most commonly used datasets in data analysis and machine learning. It contains information about passengers such as age, gender, passenger class, fare, and whether they survived the disaster. By analyzing this dataset, we can better understand the factors that influenced survival.

3. Objective

The main objectives of this internship task were:

- To clean the Titanic dataset.
- To save the cleaned file as Titanic_Cleaned.csv.
- To perform exploratory data analysis using visualizations.
- To identify trends and insights from the dataset.
- To present findings in a professional and easy-to-understand format.

4. Dataset Description

The Titanic dataset contains passenger information such as:

- Survived: Indicates whether a passenger survived or not.
- Pclass: Passenger class.
- Sex: Gender of the passenger.
- Age: Age of the passenger.
- Fare: Ticket fare paid by the passenger.

Other related features depending on the dataset version.

This dataset helps analyze how different factors such as class, gender, and age may have affected survival on the Titanic.

5. Data Cleaning Summary

Before performing analysis, the dataset was cleaned to improve accuracy and consistency. The cleaning process typically included:

- Handling missing values.
- Removing duplicate records if present.
- Converting data types if needed.
- Ensuring the dataset was ready for visualization and analysis.
- After cleaning, the dataset was saved as Titanic_Cleaned.csv.

6. Exploratory Data Analysis

EDA was performed to explore patterns and relationships in the dataset. Five visualizations were created to better understand the passenger data.

Visualization 1: Survival Count

This chart shows the total number of passengers who survived and those who did not survive.

Observation:

The number of passengers who did not survive is higher than the number of passengers who survived.

Insight:

This indicates that the dataset is slightly imbalanced, with more fatalities than survivors.

Visualization 2: Passenger Class Distribution

This chart shows how passengers are distributed across different ticket classes.

Observation:

Most passengers traveled in 3rd class, while fewer passengers were in 1st class.

Insight:

This suggests that the majority of passengers belonged to the lower ticket class.

Visualization 3: Age Distribution

This histogram displays how the ages of passengers are distributed.

Observation:

Most passengers were between 20 and 40 years old, with fewer children and elderly passengers.

Insight:

The dataset mainly contains young and middle-aged passengers.

Visualization 4: Survival by Gender

This chart compares survival outcomes between male and female passengers.

Observation:

Female passengers had a much higher survival rate than male passengers.

Insight:
Gender played an important role in survival chances during the Titanic disaster.

Visualization 5: Fare Distribution
This box plot shows the spread of ticket fares and highlights outliers.

Observation:
Most passengers paid low fares, while a few paid very high fares.

Insight:
The fare distribution is uneven, and some passengers paid significantly more than others.

7. Correlation Heatmap

An optional correlation heatmap was also created to show relationships between numerical variables.

Observation:
The heatmap helps identify correlations between different numerical features in the dataset.

Insight:
This visualization is useful for understanding which variables may be related and can help guide future analysis.

8. Key Findings

The analysis revealed several important patterns:

- More passengers died than survived.
- Most passengers were in 3rd class.
- The majority of passengers were between 20 and 40 years old.
- Female passengers had a much higher survival rate than male passengers.
- Fare values included some high outliers.

9. Conclusion

The Titanic dataset was successfully cleaned and analyzed using exploratory data analysis. The visualizations helped uncover important trends related to survival, passenger class, age, gender, and fare. The final cleaned dataset was saved as Titanic_Cleaned.csv, and the graphs provide a strong summary of the dataset's patterns.

This project improved understanding of data cleaning, visualization, and interpretation of real-world data.